

SUSHAN HARISH

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Indian Passport Holder • Full Right to Work in the UK • Open to Relocation to India

PROFESSIONAL SUMMARY

Aerospace Systems and Quality/MRO Engineer with hands-on experience across test rig development, systems integration, instrumentation, and verification & validation (V&V) in safety-critical aerospace environments. Most recently developed, integrated and commissioned fuel/oil systems environmental test rigs for **Rolls-Royce** and **Vertical Aerospace**, alongside AS9100/EASA Part 145 quality engineering experience at **Honeywell Aerospace**. MSc Aerospace Engineering (Distinction), Kingston University. Seeking Systems Engineer, Test Engineer, V&V, or Quality Engineering roles with aerospace OEMs, Tier-1 suppliers, and engineering services companies in the UK and India.

TECHNICAL PROFICIENCIES

Test & Systems Engineering: Test rig development, systems integration, subsystem (mechanical/fluid, electrical, DAQ) commissioning

Verification & Validation: Repeatability testing, baseline comparison, performance validation, comparative configuration testing, systematic troubleshooting

Instrumentation & DAQ: Flowmeters, pressure transducers, temperature probes, torque measurement — installation, calibration, scaling verification (PicoLogger, Arduino, Magtrol)

Quality & Compliance: AS9100, EASA Part 145, FMEA, RCA/RCCA, NCR management, component traceability

Manufacturing / MRO: Aircraft component inspection and assembly (CMM/SPM), production quality control, dimensional inspection

Engineering Software: Siemens NX, ANSYS, MATLAB, CFD, Excel VBA, SAP NetWeaver

PROFESSIONAL EXPERIENCE

Systems Engineer (Contract) | *Novel Engineering Consultants, UK*

Mar 2026 – Jun 2026

- Developed, integrated and commissioned a Rolls-Royce fuel/oil systems environmental test rig, working across concurrent rig programmes for Rolls-Royce and Vertical Aerospace.
- Led verification and validation activities including repeatability assessment, baseline comparison and comparative performance testing across component configurations.
- Supported environmental testing across controlled vacuum, altitude, freezer and elevated-temperature conditions.
- Delivered subsystem integration across mechanical/fluid (hydraulic/oil circuit, pressure and flow control), electrical (wiring, controller and sensor integration) and DAQ/control (PicoLogger, Arduino-driven automation) disciplines.
- Installed, calibrated and validated instrumentation including flowmeters, pressure transducers, temperature probes and torque measurement equipment.
- Diagnosed and resolved rig-level sealing and pressure-stability issues to support test safety and reliability.
- Authored test procedures, validation reports, measurement schedules and engineering handover documentation; coordinated with suppliers on test equipment procurement.
- Built Excel VBA automation converting hundreds of raw CSV test files into formatted engineering reports, reducing manual data-processing time and improving reporting consistency.

Aerospace Quality & MRO Engineer | *Honeywell Aerospace, UK*

Apr 2025 – Jan 2026

- Delivered AS9100 and EASA Part 145-compliant quality engineering within a high-volume aerospace MRO operation, supporting internal audit readiness ahead of EASA/UK CAA regulatory assessments.
- Raised and managed Non-Conformance Reports (NCRs) and supported RCCA investigations to reduce repeat defects.
- Maintained component traceability and quality documentation in SAP NetWeaver to support audit readiness.
- Inspected and assembled aircraft wheel and brake components to CMM/SPM standards and approved engineering data.
- Participated in FMEA reviews for MRO process changes to proactively identify and mitigate quality risks.
- Liaised cross-functionally with production, engineering and supplier quality teams to close out corrective actions and prevent recurrence of non-conformances.
- Contributed to internal quality management system documentation and process-improvement initiatives supporting AS9100 recertification.

Founder / Team Captain | *SAE Aero Club, St Joseph Engineering College, India*

Aug 2019 – May 2023

- Founded and led a 100-person multidisciplinary student team through the full UAV engineering lifecycle for national SAE Aero Design competitions, managing all phases from concept and design through fabrication and flight test.

- Conducted CFD simulations and aerodynamic optimisation using SolidWorks and ANSYS to maximise thrust-to-weight ratio and flight performance.
- Coached new members on design methodology, materials selection and manufacturing techniques, improving competition readiness and technical understanding across the team.
- Coordinated budgets, sponsorships and competition logistics, and presented technical progress to judges and external stakeholders at national SAE competitions.

INTERNSHIP EXPERIENCE

GE Aerospace, UK

2024

Completed a job-simulation programme advising on alternative propulsion energy sources for GE's aerospace division. Compared energy sources for propulsion with emphasis on cost, energy density and design implications, and examined bypass and compression ratios in turbofan engines. Developed a presentation outlining design constraints to support decision-making for future propulsion systems.

Hindustan Aeronautics Limited, India

2022

Aerospace Engineering Intern — supported overhaul and manufacturing operations within an aerospace OEM environment; conducted component inspection and supported corrective action processes; updated technical documentation and quality records.

Lohatec Precision Pvt Ltd, India

2022

Precision Engineering Intern — performed dimensional inspections using callipers and micrometers, verifying tolerances against engineering drawings; supported quality and process-improvement activities and component traceability.

EDUCATION

MSc Aerospace Engineering (Distinction) | Kingston University, London

2024 – 2025

Award: Best Postgraduate Project, School of Engineering. Thesis: Experimental testing and ANSYS-based simulation of Rockwool-reinforced epoxy composites, investigating wear, fatigue and failure modes to inform structural reliability and component lifespan recommendations for aerospace structures. CFD project: ONERA M6 wing benchmark validation.

BEng Mechanical Engineering (Distinction) | St Joseph Engineering College, India

2019 – 2023

Dissertation: Multi-Nut Fastening System Design and Development.

CERTIFICATIONS

- FMEA Specialist Certification (Accredited)
- Root Cause Analysis and the 8D Problem Solving Bootcamp — Udemy
- Explore Engineering Job Simulation — GE Aerospace